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**NAAC
GRADE A+**
ACCREDITED UNIVERSITY

Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No: 9

Title: Developing Flipkart Mobile Analytics Dashboard

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Experiment No: 9

Developing Flipkart Mobile Analytics KPI Dashboards

GithubLink:<https://github.com/Priyanshnolkha/Data-Analytics-For-Business-Projects/tree/main/Experiment%209%20Flipkart%20Mobile%20Analytics%20Dashboard>

Problem Statement :

You are a Business/Product Analyst tracking the mobile-phone catalog listed on Flipkart. Senior stakeholders want visibility into brand-wise pricing, discounting behaviour, storage/memory configurations, and customer rating quality across the mobile phone portfolio. This experiment creates an interactive Flipkart Mobile Analytics dashboard that helps evaluate pricing strategy, discount patterns, and product quality across brands, price categories, and storage tiers, using the Flipkart Mobiles Dataset.

Methodology :

The Flipkart_Mobiles.csv dataset (Brand, Model, Color, Memory, Storage, Rating, Selling Price, Original Price) was imported into Power BI using Power Query with Unicode (UTF-8) encoding and comma delimiter. New calculated columns (Rating Category, Rating Category Sort, Storage Sort) and DAX measures (Average Selling Price, Average Original Price, Average Rating) were created on the Flipkart_Mobiles table. All KPI cards, charts, and slicers were then arranged into a single interactive Flipkart Mobile Analytics dashboard. The dashboard includes:

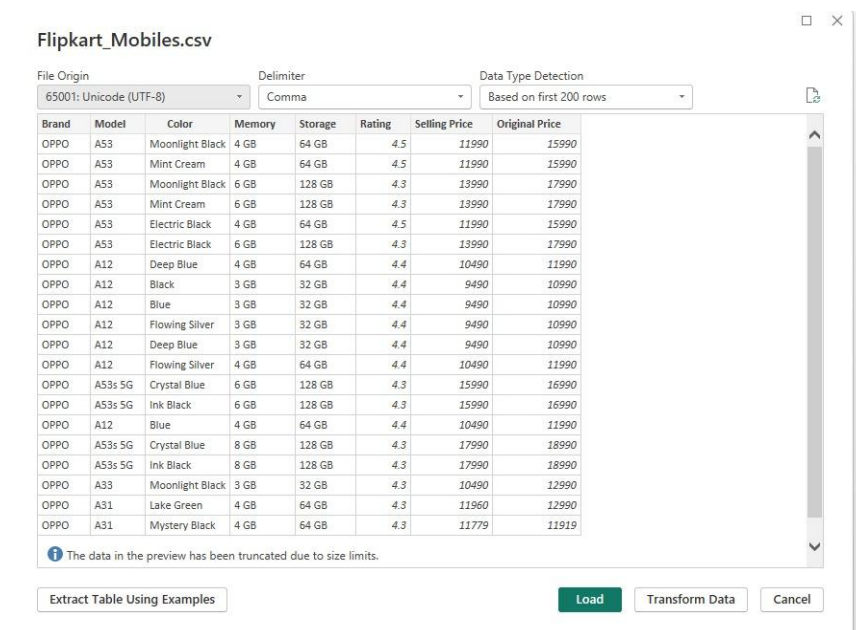
- Total Products KPI Card
- Total Brands KPI Card
- Total Models KPI Card
- Average Selling Price KPI Card
- Average Rating KPI Card
- Average Selling Price by Brand (Bar Chart)
- Original Price vs Selling Price by Brand (Combo Chart)
- Product Rating Distribution (Donut Chart)
- Average Discount % by Brand (Bar Chart)

- Products by Price Category (Column Chart)
- Storage Distribution (Column Chart)
- Interactive Slicers: Brand, Price Category, Memory, Storage, and Rating Category

The dashboard enables management to analyze brand-wise pricing and discounting, monitor product rating quality, and identify which brands, storage tiers, and price segments dominate the Flipkart mobile catalog, supporting data-driven pricing and inventory decisions.

Data Import

The Flipkart_Mobiles.csv file was loaded through Power BI's Get Data > Text/CSV connector. File origin was set to 65001: Unicode (UTF-8), the delimiter to Comma, and Data Type Detection to "Based on first 200 rows" before loading the table into the data model.



To ensure data quality before loading the table into the data model, two cleaning steps were applied in the Power Query Editor. First, Home > Remove Rows > Remove Duplicates was used across the selected columns to eliminate duplicate product listings from the catalog. Second, the Rating column contained null entries for products with no customer reviews yet; these were standardised by using Transform > Replace Values to replace all null values in the Rating column with 0, so that the column could be used safely in numeric aggregations such as Average Rating.

FileHomeTransformAdd ColumnViewToolsHelp									
Close & Apply	New Source	Recent Sources	Enter Data	Data Source Settings	Manage Parameters	Export query results	Refresh	Properties	Advanced Editor
						Query	Message	Choose Columns	Remove Columns
						Manager Columns	Remove Rows	Keep Columns	Remove Rows
						Reduct	Remove Top Rows	Remove Bottom Rows	Remove Alternate Rows
						Transform	Remove Duplicates	Remove Blank Rows	Remove Errors
						Combine	Remove rows containing duplicated values in the currently selected columns.		
						Combine Files			
						Combine			
2	OPPO	A53	Mint Cream	128 GB	4.3	23990			
3	OPPO	A53	Moonlight Black	128 GB	4.3	23990			
4	OPPO	A53	Mint Cream	128 GB	4.3	23990			
5	OPPO	A53	Electric Black	64 GB	4.3	23990			
6	OPPO	A53	Electric Black	4.3	23990				
7	OPPO	A12	Deep Blue	9 GB	4.4	20490			
8	OPPO	A12	Blue	32 GB	4.4	9490			
9	OPPO	A12	Blue	32 GB	4.4	9490			
10	OPPO	A12	Flowing Silver	32 GB	4.4	9490			
11	OPPO	A12	Deep Blue	32 GB	4.4	9490			
12	OPPO	A12	Flowing Silver	4 GB	4.4	20490			
13	OPPO	A53s 5G	Crystal Blue	6 GB	4.3	25990			
14	OPPO	A53s 5G	Ink Black	6 GB	4.3	23990			
15	OPPO	A12	Blue	4 GB	4.4	20490			
16	OPPO	A53s 5G	Crystal Blue	8 GB	4.3	17990			
17	OPPO	A53s 5G	Ink Black	8 GB	4.3	17990			
18	OPPO	A53	Moonlight Black	3 GB	4.3	23990			
19	OPPO	A31	Lake Green	4 GB	4.3	22990			
20	OPPO	A31	Mystery Black	4 GB	4.3	22779			
21	OPPO	A74 5G	Fantastic Purple	6 GB	4.3	28979			
22	OPPO	A11K	Flowing Silver	2 GB	4.2	8990			
23	OPPO	F17 Pro	Matte Black	8 GB	4.3	29990			
24	OPPO	A54	Moonlight Gold	4 GB	4.3	25990			
25	OPPO	A54	Moonlight Gold	4 GB	4.3	24990			
26	OPPO	Reno6 5G	Aurora	8 GB	4.3	29990			
27	OPPO	Reno6 5G	Stellar Black	8 GB	4.3	29990			
28	OPPO	F17 Pro	Magic Blue	8 GB	4.3	29990			
29	OPPO	A54	Starry Blue	4 GB	4.3	24990			
30	OPPO	A54	Starry Blue	4 GB	4.3	25990			
31	OPPO	A54	Moonlight Gold	6 GB	4.3	26990			
32	OPPO	A54	Starry Blue	6 GB	4.3	26990			
33	OPPO	A54	Crystal Black	4 GB	4.3	24990			
34	OPPO	F17 Pro	Metallic White	8 GB	4.3	29990			
35	OPPO	A31	Fantasy White	4 GB	4.3	22199			
36	OPPO	F17	Navy Blue	8 GB	4.3	22990			
37	OPPO	A53	Fancy Blue	6 GB	4.3	23999			
38	OPPO	A16	Pearl Blue	4 GB	0	23820			
39	OPPO	A11K	Deep Blue	2 GB	4.2	8990			
40	OPPO	A54	Crystal Black	4 GB	4.3	23990			
41	OPPO	Reno6 Pro 5G	Animal Blue	6 GB	4.4	30990			
42	OPPO	A16	CRYSTAL BLACK	4 GB	0	23990			
43									

Crystal Blue6 GB128 GB4.3

Replace Values

Replace one value with another in the selected columns.

Value To Find

null

Replace With

0

OK

Cancel

Stellar Black8 GB128 GB4.3

Magic Blue8 GB128 GB4.3

Data Visualization (Creating KPI Dashboard With Changing Values):

DAX Measures Used :

1. Average Selling Price

Screenshot Image:

```
1 Average Selling Price =
2 AVERAGE('Flipkart_Mobiles'[Selling Price])
```

2. Average Original Price

Screenshot Image:

```
1 Average Original Price =
2 AVERAGE('Flipkart_Mobiles'[Original Price])
```

3. Average Rating

Screenshot Image:

```
1 Average Rating =
2 AVERAGE('Flipkart_Mobiles'[Rating])
```

New Columns

1. Rating Category

```
1 Rating Category =
2 SWITCH(
3     TRUE(),
4     'Flipkart_Mobiles'[Rating] >= 4.5, "Excellent (4.5-5)",
5     'Flipkart_Mobiles'[Rating] >= 4.0, "Very Good (4-4.5)",
6     'Flipkart_Mobiles'[Rating] >= 3.0, "Good (3-4)",
7     'Flipkart_Mobiles'[Rating] >= 2.0, "Average (2-3)",
8     "Poor (1-2)"
9 )
```

2. Rating Category Sort

```

1 Rating Category Sort =
2 SWITCH(
3     TRUE(),
4     Flipkart_Mobiles[Rating] >= 4.5, 1,
5     Flipkart_Mobiles[Rating] >= 4.0, 2,
6     Flipkart_Mobiles[Rating] >= 3.0, 3,
7     Flipkart_Mobiles[Rating] >= 2.0, 4,
8     Flipkart_Mobiles[Rating] >= 1.0, 5,
9     99
10 )

```

3. Storage Sort

```

1 Storage Sort =
2 SWITCH(
3     TRUE(),
4     CONTAINSSTRING(UPPER('Flipkart_Mobiles'[Storage]), "32"), 1,
5     CONTAINSSTRING(UPPER('Flipkart_Mobiles'[Storage]), "64"), 2,
6     CONTAINSSTRING(UPPER('Flipkart_Mobiles'[Storage]), "128"), 3,
7     CONTAINSSTRING(UPPER('Flipkart_Mobiles'[Storage]), "256"), 4,
8     CONTAINSSTRING(UPPER('Flipkart_Mobiles'[Storage]), "512"), 5,
9     CONTAINSSTRING(UPPER('Flipkart_Mobiles'[Storage]), "1 TB"), 6,
10    CONTAINSSTRING(UPPER('Flipkart_Mobiles'[Storage]), "1TB"), 6,
11    CONTAINSSTRING(UPPER('Flipkart_Mobiles'[Storage]), "1024"), 6,
12    99
13 )

```

Dashboard Images :

Image 1st:

It displays the overall Flipkart Mobile Analytics dashboard without any filters applied (Brand, Price Category, Memory, Storage, and Rating Category slicers set to "All").

- Total Products: 3006 and Total Brands: 17, spanning 888 distinct Models — a large, multi-brand mobile catalog.
- Average Selling Price: ₹25,842 and Average Rating: 4.05, reflecting a catalog skewed toward the budget and mid-range segments but with generally well-rated products.
- The Average Selling Price by Brand chart shows Apple (₹86K) and Google Pixel (₹61K) at the premium end, while Samsung, vivo and ASUS cluster around ₹20K–24K.
- The Original Price vs Selling Price by Brand combo chart shows Apple carrying both the highest original price (₹89K) and selling price (₹62K), the widest brand-level discount gap in absolute terms.

- Product Rating Distribution shows 65.3% of products rated "Very Good (4–4.5)", 17.2% "Excellent (4.5–5)", 12.9% "Good (3–4)", and small slivers for Average and Poor — indicating strong overall product quality.
- Average Discount % by Brand is led by POCO (14.35%) and Motorola (12.54%), while Infixio and realme discount the least (~6.8%–8.7%).
- Products by Price Category peaks at Budget (<₹15K) with ~1.5K products, followed by Mid-Range (~0.8K), High-End (~0.4K) and Premium (~0.3K).
- Storage Distribution is dominated by 64GB (754) and 128GB (752) variants, with 32GB close behind (550); 256GB, 512GB and 1TB together make up a small minority.

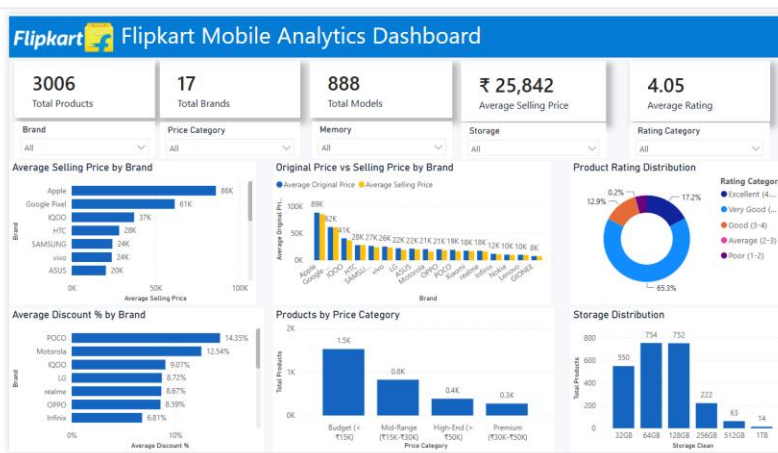


Image 2nd:

In the dashboard, after selecting "Apple" in the Brand slicer:

- All visuals are cross-filtered to show data for the Apple brand only.
- The KPI cards update to Total Products = 324, Total Brands = 1, Total Models = 24, Average Selling Price = ₹85,963, and Average Rating = 3.83.
- The Original Price vs Selling Price by Brand chart collapses to a single Apple bar (₹89K original vs ₹62K selling), confirming Apple is priced well above the portfolio average even after discounting.
- Average Discount % by Brand for Apple is negligible/near-zero compared to brands like POCO and Motorola, showing Apple relies far less on discounting than the rest of the catalog.
- Storage Distribution within Apple skews toward 128GB (71) and 256GB (90), with far fewer 32GB (21) units — consistent with Apple's higher-storage product mix.

- This confirms Apple sits in a distinct premium tier: highest price, lowest discounting, and a rating slightly below the portfolio average (3.83 vs 4.05).

Filtered by Brand = Apple

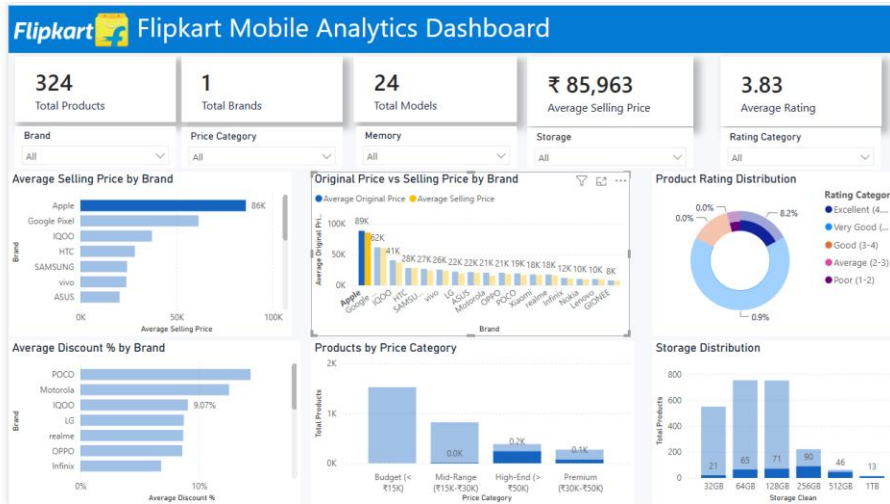


Image 3rd:

In this image, the dashboard shows another filtered view of the marketing campaign data.

- The KPI values change dynamically:
 - Total Campaigns: 40K
 - Total Clicks: 22M
 - Average Conversion Rate: 0.08
 - Average ROI: 4.99
 - Total Impressions: 220M
- The Clicks Trend Over Time chart updates according to the selected data.
- Campaign Type Distribution shows the distribution of campaign types.
- Channel-wise clicks and conversion rate are displayed.
- Average ROI is compared across different campaign types.
- Impressions and Clicks are compared using the scatter chart.

- The campaign table updates according to the selected data.

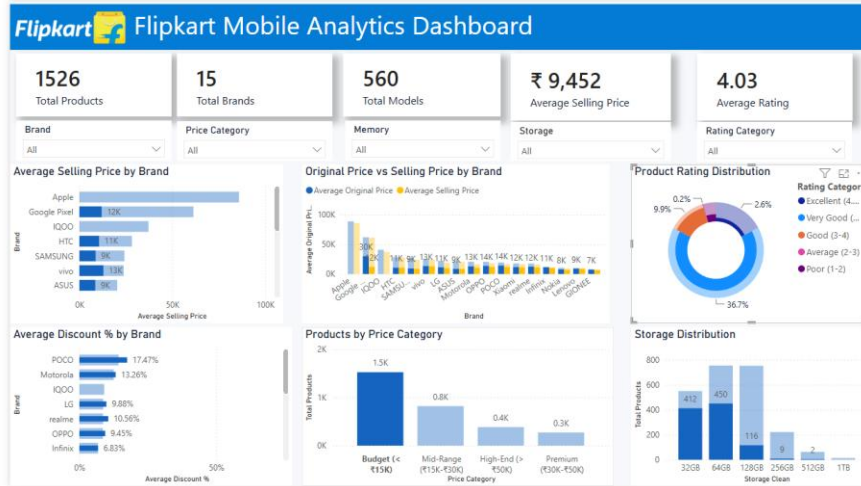


Image 4th: Filtered by Rating Category = Very Good (4–4.5)

In the dashboard, after selecting "Very Good (4–4.5)" in the Rating Category slicer:

- All visuals are cross-filtered to products rated between 4.0 and 4.5. Total Products = 1964 across all 17 brands and 619 models.
- Average Rating naturally narrows to 4.25, and Average Selling Price for this quality band comes out to ₹18,082 — between the overall and budget-only averages.
- Average Discount % by Brand within this rating band is again topped by POCO (16.87%) and Motorola (13.41%), reinforcing that discount depth is a brand-level pattern independent of product rating.
- Storage Distribution for Very-Good-rated phones is fairly balanced across 32GB (431), 64GB (536) and 128GB (521), suggesting well-rated products exist across most storage tiers rather than only at the high end.
- This view helps management identify that the "Very Good" quality band is broad-based across brands and price points, not concentrated in a single segment.

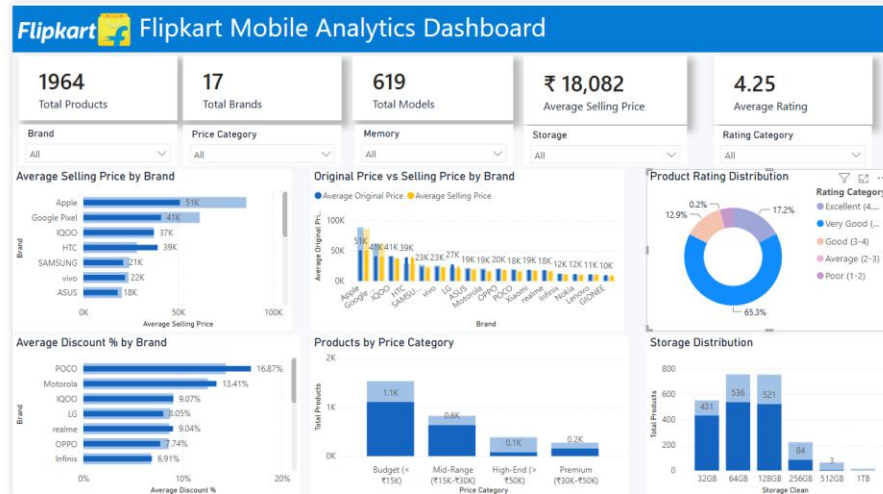


Figure 4: Flipkart Mobile Analytics Dashboard – Filtered by Rating Category = Very Good (4–4.5)

Business Insights

- The catalog comprises 3,006 products across 17 brands and 888 models, with a healthy Average Rating of 4.05 and 82.5% of products rated "Very Good" or "Excellent" — indicating a generally high-quality listed catalog.
- Apple sits in a clear premium tier of its own: Average Selling Price of ₹85,963 (over 3x the portfolio average), minimal discounting, and only 1 brand/24 models — pricing power comes from brand strength rather than volume.
- POCO and Motorola are the most aggressive discounters (14–17% average discount) across every price segment tested (overall, budget, and Very-Good-rated), suggesting discount-led positioning is a deliberate, consistent strategy for these brands rather than a one-off promotion.
- Budget (<₹15K) phones make up roughly half the catalog (1,526 of 3,006 products) and are concentrated almost entirely in 32GB and 64GB storage, confirming storage capacity is the primary lever brands use to hit budget price points.
- 64GB and 128GB are the dominant storage tiers overall (754 and 752 products respectively), while 256GB+ configurations remain a small niche (under 10% combined) — signalling where most consumer demand and inventory currently sits.
- Product quality (Very Good rating band) is broad-based across brands and storage tiers rather than concentrated in premium models, meaning affordable phones are not being rated meaningfully worse than expensive ones.
- Interactive slicers for Brand, Price Category, Memory, Storage, and Rating Category allow management to drill into specific combinations to identify which brands and configurations are driving the strongest price-to-rating value.

Conclusion

The Flipkart Mobiles dataset was successfully imported into Microsoft Power BI and transformed into an interactive Flipkart Mobile Analytics dashboard using KPI cards, a combo chart, a donut chart, and multiple bar/column charts, supported by new calculated columns (Rating Category, Rating Category Sort, Storage Sort) and DAX measures (Average Selling Price, Average Original Price, Average Rating). The dashboard effectively tracks brand-wise pricing, discounting behaviour, storage distribution, and product rating quality across the mobile phone catalog. Overall, the interactive dashboard supports data-driven pricing, discounting, and inventory-planning decisions by enabling management to monitor and evaluate the Flipkart mobile product portfolio efficiently.